# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2001-I-Q01] | Part A (Q1-Q5)

## Answer: 630

1. Write the number as $10a + b$. Neither digit can be zero. Divisibility by $a$ gives $b = ka$; divisibility by $b$ then requires $k \mid 10$. Since $ka \leq 9$, only $k = 1, 2, 5$ occur.

2. For $k = 1$, the numbers $11, 22, \ldots, 99$ sum to $495$; for $k = 2$, $12, 24, 36, 48$ sum to $120$; for $k = 5$, only $15$ works. Total $630$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2001-II-Q01] | Part A (Q1-Q5)

## Answer: 816

1. The allowable blocks are $16, 25, 36, 49, 64, 81$. They give directed digit chains $1 \to 6 \to 4 \to 9$, $2 \to 5$, $3 \to 6 \to 4 \to 9$, and $8 \to 1 \to 6 \to 4 \to 9$.

2. There is no directed cycle, so no arbitrarily long number. The unique longest chain gives $N = 81649$, whose first three digits are $816$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2002-I-Q03] | Part A (Q1-Q5)

## Answer: 025

1. Write Jane's future age as $10a + b \geq 26$. Dick's is $10b + a$, which is larger exactly when $b > a$. Each such digit pair uniquely gives $n = 10a + b - 25$ and $d = 25 + 9(b - a)$.

2. Tens digit $2$ allows $26, 27, 28, 29$, giving four. Tens digits $3$ through $8$ allow $6, 5, 4, 3, 2, 1$ choices. Their sum is $25$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2002-II-Q01] | Part A (Q1-Q5)

## Answer: 009

1. Write $x = 100a + 10b + c$, so $y = 100c + 10b + a$. Their absolute difference is $99|a - c|$.

2. Both end digits $a, c$ must be in $1$ through $9$, so $|a - c|$ takes exactly $0, 1, \ldots, 8$. Thus there are $9$ distinct values.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2003-II-Q02] | Part A (Q1-Q5)

## Answer: 120

1. At most ten digits can be used. A feasible ten-digit number beats every shorter one, so maximize the prefix lexicographically as $9876543$ if the remaining digits $0, 1, 2$ admit a suffix divisible by eight.

2. The six suffix permutations are $012, 021, 102, 120, 201, 210$; only $120$ is divisible by eight. Thus $N = 9876543120$, and the remainder is $120$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2004-I-Q01] | Part A (Q1-Q5)

## Answer: 217

1. For leading digit $a = 3, \ldots, 9$, the number is $1111a - 123$. As $1111 \equiv 1$ and $123 \equiv 12 \pmod{37}$, its remainder is $a + 25$.

2. The remainders are $28, 29, \ldots, 34$, summing to $7 \cdot 31 = 217$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2006-I-Q03] | Part A (Q1-Q5)

## Answer: 725

1. Write the number $d10^k + u$ with leading digit $d$. Then $d10^k = 28u$. Since seven does not divide a power of ten, it divides $d$, forcing $d = 7$.

2. Thus $u = 10^k/4$, requiring $k \geq 2$. The least case has $u = 25$, giving $725$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2008-II-Q04] | Part A (Q1-Q5)

## Answer: 021

1. The ordinary ternary representation is $2202101_3$. Replace each digit two by minus one with a carry to the next position, giving $2008 = 3^7 - 3^5 + 3^4 - 3^3 + 3^2 + 3^0$.

2. Balanced ternary digits are unique because reduction modulo three determines the final digit and then induction determines the rest. Exponent sum $7 + 5 + 4 + 3 + 2 + 0 = 21$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2009-I-Q01] | Part A (Q1-Q5)

## Answer: 840

1. The smallest possible leading digit is one; integral later digits force positive integer ratio at least two, giving minimum $124$. Zero cannot occur, since it would force repeated zeroes.

2. For maximum leading digit nine, the middle digit $b$ requires $b^2/9$ to be an integer digit distinct from nine and $b$. Thus $b = 3, 6$ give $931, 964$, with maximum $964$. Difference $964 - 124 = 840$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2010-II-Q01] | Part A (Q1-Q5)

## Answer: 640

1. The available digits $0, 2, 4, 6, 8$ sum to $20$. A positive multiple of $9$ using them must have even digit sum $18$; thus the used nonzero digits are exactly $4, 6, 8$, optionally with $0$.

2. The largest arrangement is $8640$. Its last two digits are divisible by $4$, so it is divisible by both $4$ and $9$, hence by $36$.

3. The remainder is $640$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2012-I-Q01] | Part A (Q1-Q5)

## Answer: 040

1. Divisibility requires $2b + a \equiv 2b + c \equiv 0 \pmod 4$. Thus $a, c$ lie together in either $\{2, 6\}$ or $\{4, 8\}$.

2. Each class gives four ordered pairs. The first requires odd $b$ and the second even $b$, each with five choices. Hence the count is $2 \cdot 4 \cdot 5 = 40$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2012-I-Q05] | Part A (Q1-Q5)

## Answer: 330

1. Adding one replaces a terminal zero followed by $k$ ones with a one followed by $k$ zeros, changing the one count by $1-k$. It preserves the required weight exactly when $k=1$.

2. Thus the smaller number ends in $01$, the larger in $10$, and the shared eleven-digit prefix has seven ones and four zeros. There are $\binom{11}{4}=330$ such prefixes and exactly one successful ordered subtraction for each.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2013-I-Q02] | Part A (Q1-Q5)

## Answer: 200

1. The last digit is zero or five, but cannot be zero because it equals the first. Thus both endpoint digits are five.

2. Choose the next two digits arbitrarily. The final middle digit must have one specified residue modulo five, with two available decimal digits. Hence the count is $10 \cdot 10 \cdot 2 = 200$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2015-II-Q03] | Part A (Q1-Q5)

## Answer: 476

1. Write the number as $17k$. Congruence to its digit sum modulo $9$ forces $17k \equiv 17 \pmod 9$, hence $k \equiv 1 \pmod 9$.

2. The first candidates are $k = 1, 10, 19, 28$, giving $17, 170, 323, 476$. The first three have digit sum $8$ and the last has digit sum $17$. Thus the least answer is $476$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2017-I-Q05] | Part A (Q1-Q5)

## Answer: 321

1. Fractional parts are below1 in both bases, so integer parts agree: $8a + b = 13b$, or $2a = 3b$. The base-eight bounds allow only $(a, b) = (3, 2), (6, 4)$.

2. The fractional equality is $(8c + d)/64 = (12b + a)/144$. The two cases respectively give $8c + d = 12$ and24. The latter requires a zero units digit, forbidden; the former yields $c = 1, d = 4$.

3. Thus the requested decimal numeral is321.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2017-II-Q04] | Part A (Q1-Q5)

## Answer: 222

1. All one- through six-digit ternary strings qualify under the bound, contributing $\sum_{k=1}^{6} 2^k = 126$, since the largest is $222222_3 = 728$.

2. Convert the bound: $2017 = 2202201_3$. A seven-digit nonzero-digit string is allowed precisely when its first two digits are11,12 or21; prefix22 would exceed the bound at the third digit. This gives $3 \cdot 2^5 = 96$.

3. Total $126 + 96 = 222$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2018-I-Q02] | Part A (Q1-Q5)

## Answer: 925

1. Place values give $196a + 14b + c = 225a + 15c + b = 222a + 37c$. The shared base-six digits satisfy $1 \le a \le 5, 0 \le c \le 5$.

2. Equating the last two forms gives $b = 22c - 3a$. The first two give $13b = 29a + 14c$. Combining yields $a = 4c$.

3. The only permitted digits are $c = 1, a = 4, b = 10$, and the value is $222(4) + 37 = 925$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2018-II-Q03] | Part A (Q1-Q5)

## Answer: 371

1. We need $b \geq 8$, $3b + 6$ a square, and $2b + 7$ a cube. The last expression is odd and below $2007$, so its positive odd cube possibilities give $b = 10, 59, 168, 361, 662$ after discarding the invalid negative case.

2. The corresponding square candidates are $36, 183, 510, 1089, 1992$. Only $36 = 6^2$ and $1089 = 33^2$ qualify, at bases ten and $361$.

3. The sum is $371$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2019-I-Q01] | Part A (Q1-Q5)

## Answer: 342

1. The geometric sum is the numeral consisting of $321$ ones followed by zero, minus $321$. Its final four-digit block changes from $1110$ to $0789$.

2. There are $318$ preceding ones. Digit sum $318+7+8+9=342$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2020-I-Q03] | Part A (Q1-Q5)

## Answer: 621

1. Place-value expansion gives $121a + 11b + c = 512 + 64b + 8c + a$, or $120a = 512 + 53b + 7c$, where $1 \leq a \leq 7$ and $0 \leq b, c \leq 7$.

2. Necessarily $a \geq 5$. At $a = 5$, solve $53b + 7c = 88$: $b = 0$ fails divisibility, $b \geq 2$ is too large, and $b = 1$ gives $c = 5$.

3. The value is $515_{11} = 621 = 1155_8$. Any $a \geq 6$ gives at least $726 > 621$, proving minimality.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2020-II-Q05] | Part A (Q1-Q5)

## Answer: 151

1. The first hexadecimal value needing a digit beyond $9$ is $10$. For an integer $k < 31$, its base-$8$ digit sum is at most $9$, while $31 = 37_8$ has digit sum $10$.

2. The least base-$4$ number with digit sum at least $31$ uses eleven digits, with leading digit $1$ and ten trailing $3$ digits. This greedy placement puts as much digit mass as possible in low places; thus it also proves minimality.

3. $N = 13333333333_4 = 2 \cdot 4^{10} - 1 = 2097151$, so the remainder is $151$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2022-I-Q02] | Part A (Q1-Q5)

## Answer: 227

1. Place value gives $100a + 10b + c = 81b + 9c + a$, or $99a = 71b + 8c$. Here $1 \leq a, b \leq 8$ and $0 \leq c \leq 8$.

2. Modulo $8$, $b \equiv 5a$. For $a = 1, \ldots, 8$, the possible positive digits $b$ are respectively $5, 2, 7, 4, 1, 6, 3, 8$. The corresponding $99a - 71b$ are $-256, 56, -200, 112, 424, 168, 480, 224$.

3. Only $56$ is in $[0, 64]$ and divisible by $8$. Thus $(a, b, c) = (2, 2, 7)$, and the answer is $227$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2023-II-Q02] | Part A (Q1-Q5)

## Answer: 585

1. A base-eight number of at most three digits is at most $511$. Any larger candidate below $1000$ has four octal digits, beginning and ending with $1$.

2. Its form is $\overline{1bb1}_8 = 513 + 72b$, with $b = 0, \ldots, 6$. The candidates are $513, 585, 657, 729, 801, 873, 945$.

3. Only $585$ is a decimal palindrome; it exceeds all three-digit octal numbers and is therefore the maximum. Indeed $585 = 1111_8$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2024-II-Q03] | Part A (Q1-Q5)

## Answer: 045

1. Write rows as $a, b, c$ and $d, e, f$. The row sum forces $a + d = b + e = c + f = 9$: the units sum cannot be $19$, so no carries occur, and the same argument repeats.

2. Let $S = a + b + c$. The column sum is $10S + (27 - S) = 99$, hence $S = 8$. Each nonnegative triple summing to $8$ is valid and fixes the bottom digits.

3. No digit upper bound is active, so stars and bars gives $\binom{10}{2} = 45$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2025-I-Q01] | Part A (Q1-Q5)

## Answer: 070

1. The divisibility is $(b+7) \mid (9b+7)$, equivalent to $(b+7) \mid 56$ by subtraction of $9(b+7)$.

2. Since $b+7 > 16$, it is $28$ or $56$. Thus $b = 21, 49$, whose sum is $70$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2026-I-Q02] | Part A (Q1-Q5)

## Answer: 062

1. An even-length palindrome has even digit sum, so the middle digit is one of $1, 3, 5, 7, 9$. The left half then has digit sum $s = 6, 5, 4, 3, 2$, respectively, and determines the whole palindrome.

2. Since $s \leq 6$, nonzero-digit strings of sum $s$ are exactly compositions of $s$, numbering $2^{s-1}$. The sum is $32 + 16 + 8 + 4 + 2 = 62$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2026-II-Q04] | Part A (Q1-Q5)

## Answer: 279

1. Every one-digit number works. For a number with at least two digits, base $b < 10$ strictly lowers its value because the positive leading-digit weight decreases. Thus a multidigit number works exactly when it contains digit nine.

2. There are nine one-digit numbers, $90 - 8 \cdot 9 = 18$ two-digit numbers containing nine, and $900 - 8 \cdot 9^2 = 252$ three-digit ones. Total $9 + 18 + 252 = 279$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2001-I-Q08] | Part B (Q6-Q10)

## Answer: 315

1. For digits $d_i \in \{0, \ldots, 6\}$, the condition is $\sum d_i(10^i - 2 \cdot 7^i) = 0$. The coefficients for $i = 0, 1, 2, 3$ are $-1, -4, 2, 314$; subsequent coefficients are even larger positive numbers.

2. The total negative contribution is at most $30$, so no position $i \geq 3$ can be nonzero. Thus $2d_2 = 4d_1 + d_0$.

3. To maximize $49d_2 + 7d_1 + d_0$, choose $d_2 = 6$ and then $d_1 = 3, d_0 = 0$. The resulting $630_7 = 315$ indeed reads as $630 = 2N$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2006-II-Q07] | Part B (Q6-Q10)

## Answer: 738

1. There are $999$ positive pairs. Exactly $99$ have both units digits zero: $a = 10u, b = 10v, u + v = 100$.

2. If units digits are nonzero, a forbidden number must have form $h0u$ with $h, u = 1, \ldots, 9$. There are $81$ choices for either member; its complement has tens digit nine and cannot also have this form. Hence these exclude $162$ additional pairs.

3. Valid count $999 - 99 - 162 = 738$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2012-II-Q07] | Part B (Q6-Q10)

## Answer: 032

1. Below $2^{12}$ are $\binom{12}{8} = 495$ such numbers, and below $2^{13}$ are $\binom{13}{8} = 1287$. The desired one is thirteen-bit. There are $\binom{11}{6} = 462$ beginning $11$, the largest group, so its rank within that group is $462 - (1287 - 1000) = 175$.

2. The initial groups $1100$, $11010$, $110110$ have counts $\binom{9}{6} = 84$, $\binom{8}{5} = 56$, $\binom{7}{4} = 35$, whose sum is $175$. Thus take the greatest member of the final group, placing four remaining ones as far left as possible.

3. The binary string is $1101101111000_2 = 7032$, leaving remainder $32$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2015-I-Q08] | Part B (Q6-Q10)

## Answer: 695

1. Numbers with fewer than three digits have digit sum at most $18$. In addition, $s(864) = 18$, and each carry reduces the naive sum by $9$, so exactly two carries must occur.

2. For a three-digit candidate $n = 100a + 10b + c$, $a + b + c = 20$ forces $a \geq 2$, guaranteeing a hundreds carry. If units carry but tens do not, $c \geq 6, b \leq 2$, so the minimum possible candidate is $929$.

3. If tens carry but units do not, $c \leq 5$, so $a \geq 20 - 9 - 5 = 6$. The smallest realization is $(a, b, c) = (6, 9, 5)$, giving $695$. Indeed $695 + 864 = 1559$ and both digit sums are $20$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2016-I-Q08] | Part B (Q6-Q10)

## Answer: 162

1. Let the hundreds-column sum be $H$ and units-column sum be $U$. Since all digits total $45$, the overall sum is $100H + 10(45 - H - U) + U = 90H + 450 - 9U$.

2. Three distinct units digits sum between $6$ and $24$; ending in $0$ requires $U = 10$ or $20$. Since $H \geq 6$, the respective lower bounds are $900$ and $810$. The latter is attainable only with hundreds digits $\{1, 2, 3\}$ and $U = 20$.

3. From remaining digits $4, 5, 6, 7, 8, 9$, the units triples summing to $20$ are $\{4, 7, 9\}, \{5, 6, 9\}, \{5, 7, 8\}$. Each permits $(3!)^3$ assignments to the three columns, so $n = 3 \cdot 216 = 648$.

4. Thus $m = 810$ and $|m - n| = 162$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2000-II-Q14] | Part C (Q11-Q15)

## Answer: 495

1. The identity $(i+1)! - i! = i \cdot i!$ gives $a! - b! = \sum_{i=b}^{a-1} i \cdot i!$. Group $k$ as $16! + \sum_{h=1}^{62}[(32h+16)! - (32h)!]$.

2. Thus $f_{16} = 1$ and, in each interval $32h \le i \le 32h+15$, $f_i = i$; other digits are zero. These are legal digits on disjoint intervals, so no carrying is needed.

3. Each sixteen-digit interval starts at an even index and contributes eight pairs $-i + (i+1) = 1$. The initial digit contributes $-1$. Total $62 \cdot 8 - 1 = 495$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2003-I-Q13] | Part C (Q11-Q15)

## Answer: 155

1. A length-$(d+1)$ binary number starts with one. If its remaining $d$ digits contain $j$ ones, the required inequality is $j \geq d/2$. The count through $2047$ is therefore the upper halves of binomial rows $0$ through $10$.

2. By symmetry this count is $[2047 + (1 + 2 + 6 + 20 + 70 + 252)]/2 = 1199$.

3. The $44$ integers $2004$ through $2047$ all have prefix $11111$ and at least one more one, so all were included. Thus $N = 1155$, with remainder $155$.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2014-II-Q15] | Part C (Q11-Q15)

## Answer: 149

1. Index primes as $\rho_0 = 2, \rho_1 = 3, \rho_2 = 5, \ldots$. The invariant is $x_n = \prod_i \rho_i^{d_i}$, where $d_i$ are the binary digits of $n$. It holds at zero, giving the empty product.

2. For the first missing prime $\rho_j$, all preceding primes occur once. The update removes them and inserts $\rho_j$, leaving higher primes unchanged. This is exactly a binary carry through the trailing one-bits, so induction proves the invariant and squarefreeness.

3. Now $2090 = 2 \cdot 5 \cdot 11 \cdot 19 = \rho_0\rho_2\rho_4\rho_7$, so its unique binary index is $2^0 + 2^2 + 2^4 + 2^7 = 149$. Uniqueness of prime and binary representations proves this is the first occurrence.

Source / 来源

# Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2024-II-Q14] | Part C (Q11-Q15)

## Answer: 211

1. Write the digits as $x, y$ and their sum as $s$. Then $s^2 = bx + y = (b-1)x + s$, so $(b-1) \mid s(s-1)$. Conversely every solution $2 \le s \le b-1$ gives digits $x = s(s-1)/(b-1)$ and $y = s(b-s)/(b-1)$, which satisfy the digit bounds.

2. Put $M = b - 1$. Since consecutive integers are coprime, each prime-power factor of $M$ must divide either $s$ or $s-1$. The Chinese remainder theorem gives $2^{\omega(M)}$ residue classes; excluding $s = 1$ leaves $2^{\omega(M)} - 1$ special integers.

3. This exceeds ten exactly when $\omega(M) \ge 4$. The least such $M$ is $2 \cdot 3 \cdot 5 \cdot 7 = 210$, hence $b = 211$.

Source / 来源